

Grade 8
Unit 4
Lesson 1 Practice

Name Answer key
Date _____
Period _____

Write an equivalent exponential expression by applying the laws of exponents.

1. $(2^6)^2 = \underline{2^{12}}$

2. $11^1 \cdot 11^7 = \underline{11^8}$

3. $\left(\frac{-3}{10}\right)^4 = \frac{(-3)^4}{10^4}$

4. $(6 \cdot -8)^3 = \underline{6^3 \cdot (-8)^3}$

Rewrite each expression as an equivalent exponential expression using positive exponents.

5. $\frac{1}{-5^{-5}} = \underline{-5^5}$

6. $(-3)^{-9} = \frac{1}{(-3)^9}$

7. $3^{-4} = \frac{1}{3^4}$

8. $\frac{3^{-7}}{12^{-1}} = \frac{12}{3^7}$

9. For each expression, rewrite the exponential expression applying the negative exponent law, writing in expanded form, applying the product of powers exponent law and evaluate.

Exponential Expression	Apply Negative Exponent Law	Expanded Form	Apply Product of Powers Exponent Law	Value
$(-3)^2 \cdot (-3)^{-5}$	$(-3)^2 \cdot \frac{1}{(-3)^5}$	$\frac{(-3)(-3)}{(-3)(-3)(-3)(-3)(-3)}$	$(-3)^{2+(-5)}$	$= \frac{1}{-3^3}$ or $\frac{1}{-27}$
$2^8 \cdot 2^{-4}$	$2^8 \cdot \frac{1}{2^4}$	$\frac{2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2}{2 \cdot 2 \cdot 2 \cdot 2}$	$2^{8+(-4)}$	2^4 or 16
$9^9 \cdot 9^{-9}$	$9^9 \cdot \frac{1}{9^9}$	$\frac{9 \cdot 9 \cdot 9 \cdot 9 \cdot 9 \cdot 9 \cdot 9 \cdot 9 \cdot 9}{9 \cdot 9 \cdot 9 \cdot 9 \cdot 9 \cdot 9 \cdot 9 \cdot 9 \cdot 9}$	$9^{9+(-9)}$	$9^0 = 1$

10. For each expression, rewrite the exponential expression applying the negative exponent law, writing in expanded form, applying the quotient of powers exponent law and evaluate.

Exponential Expression	Apply Negative Exponent Law	Expanded Form	Apply Quotient of Powers Exponent Law	Value
$(-5)^{-6} \div (-5)^{-4}$	$\frac{1}{(-5)^6} \div \frac{1}{-5^4}$ or $\frac{1}{(-5)^6} \cdot (-5)^4$	$\frac{-5 \cdot -5 \cdot -5 \cdot -5}{-5 \cdot -5 \cdot -5 \cdot -5 \cdot -5 \cdot -5}$	$(-5)^{-6-(-4)}$	$= \frac{1}{(-5)^2}$ or $\frac{1}{25}$
$2^0 \div 2^{-3}$	$1 \div \frac{1}{2^3}$ or $1 \cdot \frac{1}{2^3}$	$1 \cdot 2 \cdot 2 \cdot 2$	$2^{0-(-3)}$	$2^3 = 8$
$3^{-7} \div 3^5$	$\frac{1}{3^7 \cdot 3^5}$	$\frac{1}{3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3}$	$3^{(-7-5)}$	$\frac{1}{3^{12}} = \frac{1}{531,441}$